

\$15.10

	A	B	C	D	E	F	G
1 Input data							
2 Initial inventory		5000					
3 Holding cost as % of prod cost		5%					
5 Month	1	2	3	4	5	6	
6 Production cost/unit	\$12.50	\$12.55	\$12.70	\$12.80	\$12.85	\$12.95	
7 Holding cost/unit	\$0.625	\$0.628	\$0.635	\$0.640	\$0.643	\$0.648	
8 Production plan (all quantities are in 100s of footballs)							
9 Month	1	2	3	4	5	6	
10 Units produced	5000	20000	30000	30000	25000	10000	
11	<=	<=	<=	<=	<=	<=	
12 Production capacity	30000	30000	30000	30000	30000	30000	
16 Demand	10000	15000	30000	35000	25000	10000	
18 Ending inventory (>=0)	0	5000	5000	0	0	0	
19	<=	<=	<=	<=	<=	<=	
20 Storage capacity	10000	10000	10000	10000	10000	10000	
22 Summary of costs (all costs are in hundreds of dollars)							
23 Month	1	2	3	4	5	6	Totals
24 Holding cost	0	3137.5	3175	0	0	0	6,312.50
25 Production cost	62500	251000	381000	384000	321250	129500	1,539,250.00
26							1,535,562.50

Input parameters

- n = number of decision periods
- p_t = unit production cost in period t
- h_t = unit holding cost in period t
- d_t = demand in period t
- κ = warehouse capacity
- u = production capacity
- l_0 = initial inventory

Decision variable

- x_t = production quantity in period t
- I_t = inventory at end of period t

Minimize

$$\sum_{t=1}^n (p_t x_t + h_t I_t)$$

Subject to the Constraints:

$$I_t = I_{t-1} + x_t - d_t \text{ for } t = 2, \dots, n$$

$$I_1 = I_0 + x_1 - d_1$$

$$0 \leq I_t \leq \kappa \text{ for } t = 1, 2, \dots, n$$

$$0 \leq x_t \leq u \text{ for } t = 1, 2, \dots, n$$

	A	B	C	D	E	F
1	x1	x2	x3	x4	x5	
2 fix cost	1500	1200	1600	1500	1600	
3 labor hour	2	1	6	4	8	
4 cloth use	3	2.5	4	4.5	5.5	
5 sell price	35	40	65	70	110	
6 cost price	20	10	25	30	35	
7 revenue	15	30	40	40	75	
9	x1	x2	x3	x4	x5	
10 produce?	0	1	0	0	1	
11 # produce	0	965.52	0	0	379.31	
12	<=	<=	<=	<=	<=	
13	0	1000000	0	0	1000000	

Decision variable

- Let $y_j = \begin{cases} 1 & \text{we produce type } j \\ 0 & \text{otherwise} \end{cases}$
- Let x_j = Number of types- j units to be produced; $j = 1, 2, \dots, 5$

Objective function

$$\text{Max } [(35 - 20)x_1 - 1500y_1] + \dots + [(110 - 35)x_5 - 1600y_5]$$

Constraints

$$2x_1 + \dots + 8x_5 \leq 4000$$

$$3x_1 + \dots + 5.5x_5 \leq 4500$$

$$x_j \geq 0; j = 1, 2, \dots, 5$$

$$y_j \in \{0, 1\}; j = 1, 2, \dots, 5$$

Set Objective: \$1\$7

To: ☒ Max ☐ Min ☐ Value Of:

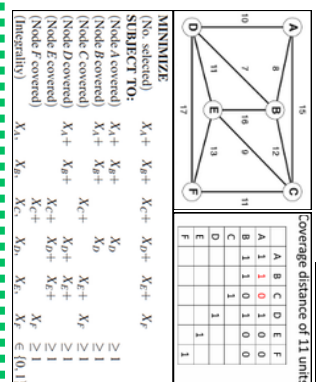
By Changing Variable Cells: \$B\$10:\$F\$11

Subject to the Constraints:

\$B\$10:\$F\$10 = binary

\$B\$11:\$F\$11 <= \$B\$13:\$F\$13

\$I\$1:\$I\$2 <= \$K\$1:\$K\$2



	A	B	C	D	E	F	G	H	I	J
1	0	0	0	1	0	1	2			
2	1	1	0	1	0	0	1	>=	1	
3	1	1	0	1	0	0	1	>=	1	
4	0	0	1	0	1	1	1	>=	1	
5	1	1	0	1	1	1	0	>=	1	
6	0	0	1	1	1	1	0	>=	1	
7	0	0	1	1	1	1	0	>=	1	
8	0	0	1	1	1	1	0	>=	1	
9	1	0	0	10	0	0	1	>=	1	
10	1	1	1	1	1	1	1	>=	1	
11	1	1	1	1	1	1	1	>=	1	
12	>=	(a1 * x1)	>=	1	site	cover	อย่างน้อย	1	ๆ	(ถ้าไม่ใส่ xj จะเลือกเป็น 0 ทั้งหมด)

Set Objective: \$H\$2

☐ Max ☒ Min

By Changing Variable Cells: \$B\$2:\$G\$2

Subject to the Constraints:

\$B\$2:\$G\$2 = binary

\$H\$4:\$H\$9 >= \$J\$4:\$J\$9

Day of Week	Minimum Number of Employees Required
Monday	17
Tuesday	13
Wednesday	15
Thursday	19
Friday	14
Saturday	16
Sunday	11

Input variables Minimum required number of workers each day

Decision variables (changing cells) Number of employees working each of the five-day shifts (defined by their first day of work)

Objective cell Total number of employees on the payroll

Other calculated variables Number of employees working each day

Constraints Employees working >= Employees required

	A	B	C	D	E	F	G	H	I	J	K
1	shift start	Mon	Tue	Wed	Thurs	Fri	Sat	Sun			
2	decis var	7	1	3	6	1	5	0	23		
4	Mon	1	1	1	1	1	0	0	18	>=	17
5	Tue	0	1	1	1	1	1	0	16	>=	13
6	Wed	0	0	1	1	1	1	1	15	>=	15
7	Thurs	1	0	0	1	1	1	1	19	>=	19
8	Fri	1	1	0	0	1	1	1	14	>=	14
9	Sat	1	1	1	0	0	1	1	16	>=	16
10	Sun	1	1	1	1	0	0	1	17	>=	11
12	fix cost	1	1	1	1	1	1	1			

Set Objective: \$I\$7

To: ☒ Max ☐ Min ☐ Value Of:

By Changing Variable Cells: \$B\$10:\$F\$11

Subject to the Constraints:

\$B\$10:\$F\$10 = binary

\$B\$11:\$F\$11 <= \$B\$13:\$F\$13

\$I\$1:\$I\$2 <= \$K\$1:\$K\$2

Problem statement

Revisit the multi-period production problem over an infinite planning horizon.

- Demand rate is constant at 20,000 units/month or $d = (12)(20000) = 240,000$ units/yr. $\rightarrow d = 240000$
- Fixed setup cost $K = 15000$ THB/batch $\rightarrow K = 15000$
- Unit holding cost 0.64 THB/unit/month or $h = (12)(0.64) = 7.68$ THB/unit/year. $\rightarrow h = 7.68$
- What batch size Q minimizes the average annual cost?

Model formulation

We formulate the classical economic order quantity (EOQ) model as a simple nonlinear optimization problem with a single decision variable.

Single item

demand rate, d (unit/year)	240000
Setup cost, K (THB/order)	15000
Unit holding cost, h (THB/unit/year)	7.68
Q	60000
Order frequency (order/year), d/Q	4
Avg inv. (unit), $Q/2$	30000
Avg setup cost (THB/year), Kd/Q	60000
Avg holding cost (THB/year), $hQ/2$	230400
Avg total cost (THB/year), $Kd/Q + hQ/2$	290400

	A	B	C	D	E	F	G
1	customer	lat	long	#shipment	x0-xi	y0-yi	distance
2	1	5	10	200	4.31	-4.97	6.58
3	2	10	5	150	-0.69	0.03	0.69
4	3	0	12	200	9.31	-6.97	11.63
5	4	12	0	300	-2.69	5.03	5.70
6	X	9.31	5.03				
7	total distance	5456.54					

Set Objective: \$B\$7

☐ Max ☒ Min

By Changing Variable Cells: \$B\$6:\$C\$6

Subject to the Constraints:

\$B\$6:\$C\$6 >= \$E\$5:\$E\$5

\$B\$6:\$C\$6 >= \$F\$5:\$F\$5

\$B\$6:\$C\$6 >= \$G\$5:\$G\$5

\$B\$6:\$C\$6 >= \$H\$5:\$H\$5

\$B\$6:\$C\$6 >= \$I\$5:\$I\$5

\$B\$6:\$C\$6 >= \$J\$5:\$J\$5

\$B\$6:\$C\$6 >= \$K\$5:\$K\$5

\$B\$6:\$C\$6 >= \$L\$5:\$L\$5

\$B\$6:\$C\$6 >= \$M\$5:\$M\$5

\$B\$6:\$C\$6 >= \$N\$5:\$N\$5

\$B\$6:\$C\$6 >= \$O\$5:\$O\$5

\$B\$6:\$C\$6 >= \$P\$5:\$P\$5

\$B\$6:\$C\$6 >= \$Q\$5:\$Q\$5

\$B\$6:\$C\$6 >= \$R\$5:\$R\$5

\$B\$6:\$C\$6 >= \$S\$5:\$S\$5

\$B\$6:\$C\$6 >= \$T\$5:\$T\$5

\$B\$6:\$C\$6 >= \$U\$5:\$U\$5

\$B\$6:\$C\$6 >= \$V\$5:\$V\$5

\$B\$6:\$C\$6 >= \$W\$5:\$W\$5

\$B\$6:\$C\$6 >= \$X\$5:\$X\$5

\$B\$6:\$C\$6 >= \$Y\$5:\$Y\$5

\$B\$6:\$C\$6 >= \$Z\$5:\$Z\$5

\$B\$6:\$C\$6 >= \$AA\$5:\$AA\$5

\$B\$6:\$C\$6 >= \$AB\$5:\$AB\$5

\$B\$6:\$C\$6 >= \$AC\$5:\$AC\$5

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